DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 23: Geometric Brownian Motion

The Model for Stock Prices · Solving the SDE · The $-\sigma^2/2$ Puzzle

September 25, 2026

Today's Agenda

1. Why not just use Brownian motion?

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3. Solving the SDE with Itô's Lemma

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3. Solving the SDE with Itô's Lemma
4. The $-\sigma^2/2$ puzzle

Learning Objectives

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- Write the GBM SDE and interpret μ, σ.
- Solve it using Itô's Lemma.
- Explain what $-\sigma^2/2$ is and why it appears.

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- **It can go negative.** $W_t \sim \mathcal{N}(0, t)$ takes negative values. A share price cannot.
- **It moves in absolute terms.** Brownian increments are the same size regardless of level – but Rs. 10 means something very different for a Rs. 50 stock than a Rs. 5000 stock.

Populations Grow by Percentages, Not Headcounts

The Right Random Quantity

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Geometric Brownian Motion

Definition

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t \qquad \Longleftrightarrow \qquad \frac{\mathrm{d}S_t}{S_t} = \mu \,\mathrm{d}t + \sigma \,\mathrm{d}W_t$$

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The proportional return over $\mathrm{d}t$ *has mean* $\mu \,\mathrm{d}t$ *and fluctuation* $\sigma \,\mathrm{d}W_t$. Both terms scale with S_t – "percentages, not headcounts." And as $S_t \to 0$ both vanish, so the process **cannot cross zero**.

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Everything in S_t cancels:

Key Takeaway

$$\mathrm{d}(\ln S_t) = \left(\mu - \tfrac{1}{2}\sigma^2\right)\mathrm{d}t + \sigma\,\mathrm{d}W_t$$

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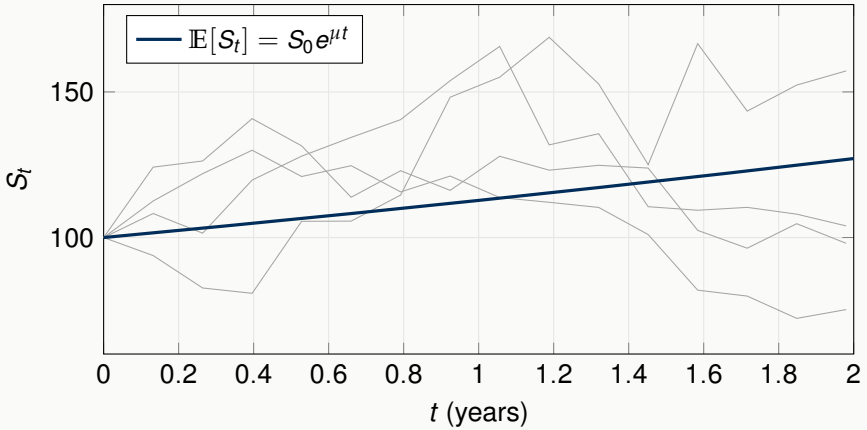
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The GBM Solution

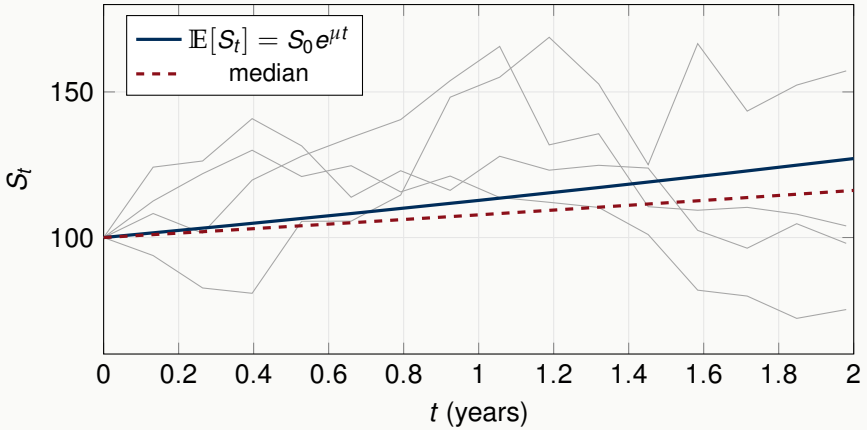
$$S_t = S_0 \exp\left[\left(\mu - \tfrac{\sigma^2}{2}\right) t + \sigma W_t\right]$$

$$\ln \frac{S_t}{S_0} \sim \mathcal{N}\left(\left(\mu - \tfrac{\sigma^2}{2}\right) t,\ \sigma^2 t\right)$$

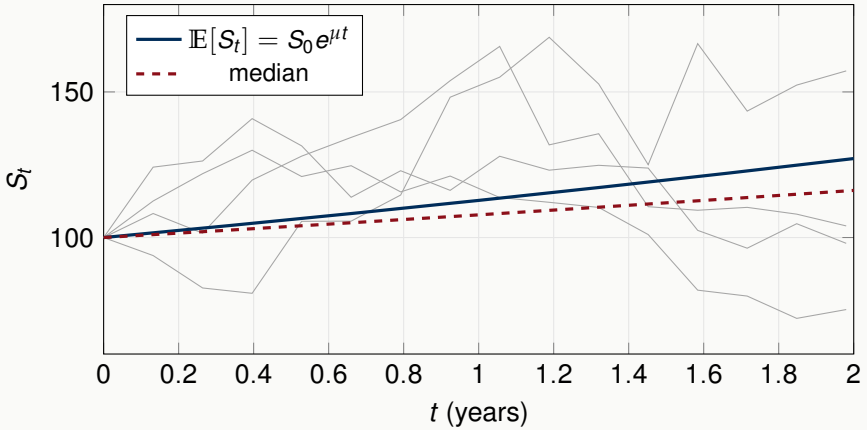
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All paths stay positive; the mean sits *above* the median.

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Both – They Answer Different Questions

$$\mathbb{E}[S_t] = S_0 e^{\mu t} \qquad \text{but} \qquad \mathbb{E}\left[\ln \frac{S_t}{S_0}\right] = \left(\mu - \tfrac{\sigma^2}{2}\right) t$$

μ: growth of the **average price**.
$\mu - \sigma^2/2$: growth of the **typical (median) path**.

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S_0=100, μ=12%, σ=30%, T=2, 500,000 paths

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Fewer than half of outcomes beat the mean – it's dragged up by a minority of large ones.

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Writing $S_t = S_0 e^{\mu t + \sigma W_t}$ does *not* give mean growth μ. It gives $\mu + \sigma^2/2$. In our example the simulated mean rises 127.12 $\to$ 139.24, inflated by $e^{\sigma^2 T/2} = 1.094$.

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Writing $S_t = S_0 e^{\mu t + \sigma W_t}$ does *not* give mean growth μ. It gives $\mu + \sigma^2/2$. In our example the simulated mean rises 127.12 $\to$ 139.24, inflated by $e^{\sigma^2 T/2} = 1.094$. A real and common bug: prices drift systematically high, and every derivative priced off them is wrong.

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- **Solution**: $S_t = S_0 \exp[(\mu - \sigma^2/2)t + \sigma W_t]$ – log-returns are normal.
- μ governs the mean; $\mu - \sigma^2/2$ governs the median.

Check Your Understanding

Discuss (2 minutes)

1. $S_0 = 50$, $\mu = 8\%$, $\sigma = 20\%$: find $\mathbb{E}[S_3]$ and median S_3.

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2. What happens to the median path when $\mu = \sigma^2/2$?
3. Apply Itô to $S_0 \exp[(\mu - \sigma^2/2)t + \sigma W_t]$ to recover the SDE.

Looking Ahead

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Lognormal Prices – Lecture 24